

Ryan Cho

Senior Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Software engineer with seven years delivering production full stack systems across finance and SaaS. Owns end-to-end design of Python backend services and responsive React applications, from REST API contracts and PostgreSQL data models through Dockerized deployment and cloud operations. Combines rigorous automated testing, performance tuning, and security-conscious development with mentoring and architectural guidance in distributed remote teams.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Led design and delivery of **Python FastAPI microservices** powering portfolio analytics for institutional clients, serving **50K+ daily API requests** with sub-200ms p95 latency.
- Built **React and TypeScript dashboards** with accessible, responsive UI patterns, reducing report generation time for analysts by **35%** through optimized client-side caching and lazy loading.
- Architected **RESTful API contracts** and OpenAPI documentation adopted across three product teams, standardizing integration patterns and cutting partner onboarding from weeks to days.
- Designed **PostgreSQL schemas and SQLAlchemy models** for multi-tenant SaaS data isolation, improving query performance **40%** via indexing strategy and read-replica routing.
- Established **unit and integration test suites** with pytest and Jest achieving **85%+ coverage** on critical payment and reconciliation workflows, gating releases through GitHub Actions CI pipelines.
- Containerized application stacks with **Docker** and deployed to **AWS** (ECS, RDS, S3), implementing blue-green releases and infrastructure-as-code templates for repeatable environments.
- Partnered with product and design stakeholders to translate compliance requirements into **secure authentication flows** (OAuth2/JWT), addressing OWASP guidelines across frontend and backend surfaces.
- Mentored **two junior engineers** on full stack best practices, code review standards, and API design, elevating team delivery velocity and reducing production defect rate **25%**.
- Diagnosed and resolved **production performance bottlenecks** through query tuning, Redis caching layers, and distributed tracing, restoring SLA compliance during peak trading windows.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Python and Django backend services** supporting trade settlement workflows, processing **100K+ transactions daily** across cross-border payment rails with audited error handling.
- Implemented **React.js interfaces** for internal operations teams, replacing legacy forms with modular component libraries and cutting manual data entry errors **30%**.
- Owned **MySQL and PostgreSQL integration** via Django ORM, optimizing batch reconciliation jobs that reduced end-of-day processing windows by **2 hours**.
- Designed and consumed **RESTful APIs** connecting core banking systems with third-party vendors, documenting schemas and versioning strategies for downstream consumers.
- Authored **automated integration tests** and participated in peer code reviews, improving release confidence for monthly production deployments in a regulated environment.
- Collaborated with business analysts and infrastructure teams to **containerize legacy applications with Docker**, enabling consistent staging environments and smoother UAT cycles.
- Supported **cloud migration initiatives on AWS**, assisting with EC2 workload cutovers, RDS provisioning, and secrets management aligned to enterprise security policies.

- Contributed to **agile sprint ceremonies** and technical runbooks, documenting incident response procedures that shortened mean time to recovery for P1 outages.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **JavaScript and React features** for inventory tracking workflows, integrating REST endpoints and improving warehouse staff task completion speed **20%**.
- Implemented **Flask API endpoints** for mobile companion apps, validating payloads and persisting records to **PostgreSQL** with SQLAlchemy ORM patterns.
- Wrote **unit tests with pytest** and frontend snapshot tests, catching regressions before weekly releases in a fast-paced startup environment.
- Assisted senior engineers with **Docker Compose** local development setups and GitLab CI configuration, streamlining onboarding for new team members.
- Refactored legacy **HTML/CSS templates** into reusable React components, improving maintainability and visual consistency across customer-facing pages.
- Participated in **code reviews and sprint planning**, delivering scoped features independently while maintaining clear async communication in a remote team.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **responsive web pages** with HTML5, CSS3, and JavaScript for nonprofit fundraising campaigns, improving mobile donation completion rates.
- Integrated **third-party REST APIs** for payment processing and donor management, handling error states and validation for reliable checkout flows.
- Maintained **content management workflows** and cross-browser compatibility testing, ensuring consistent experiences across donor-facing properties.
- Collaborated with stakeholders to translate **design mockups into production UI**, iterating quickly on feedback during a remote volunteer engagement.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React.js, TypeScript, JavaScript, HTML5, CSS3, Vue.js, Next.js, Vite, Webpack, responsive and accessible UI design

Backend: Python, FastAPI, Django, Flask, RESTful API design, GraphQL, SQLAlchemy, OAuth2, JWT, microservices

Databases: PostgreSQL, MySQL, Redis, Django ORM, SQLAlchemy, query optimization, caching strategies

Cloud & DevOps: AWS (ECS, RDS, S3, EC2), Docker, Kubernetes, GitHub Actions, GitLab CI, CI/CD pipelines, infrastructure as code

Monitoring & Tools: Distributed tracing, structured logging, Prometheus, Grafana, pytest, Jest, Git, OpenAPI/Swagger

Workflow & PM: Agile, Scrum, Jira, technical documentation, code reviews, mentoring, cross-functional stakeholder collaboration